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Beyond market stress: Incremental long-term information in geopolitical tension for gold volatility

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ABSTRACT

Against the backdrop of a reshaping global economic landscape and escalating U.S.-China geopolitical tensions, gold's significance as a traditional safe-haven asset has become increasingly prominent. This study constructs and applies the GJR-GARCH-MIDAS model based on the U.S.-China Tension Index (UCT) to systematically analyze the impact of geopolitical tensions on the long-term volatility of daily log returns in the gold spot market. Empirical results indicate that UCT has a significant positive influence on gold's long-term volatility, suggesting that heightened geopolitical tensions substantially increase long-term volatility in the gold market. This conclusion remains robust even after controlling for monthly realized volatility. Furthermore, decomposing the UCT into positive shocks (UCT^+) and negative shocks (UCT^-) reveals that both escalations and de-escalations in U.S.-China relations significantly amplify gold's long-term volatility, with UCT^+ exerting a notably stronger effect than UCT^-. Finally, out-of-sample forecasting tests demonstrate that models incorporating both realized volatility (RV) and UCT^+/UCT^- exhibit superior predictive power and robustness in forecasting gold's long-term volatility.

1. Introduction

As the global economic landscape continues to undergo restructuring, geopolitical tensions are emerging as a significant systemic factor influencing financial market volatility (Salisu et al., 2022). The pricing mechanisms in international capital markets have shifted from being driven by economic fundamentals to being shaped by the combined forces of political events, strategic competition, and policy expectations (Caldara and Iacoviello, 2022). As the most influential bilateral relationship today, the uncertainty surrounding U.S.-China relations is profoundly reshaping the global risk pricing system. The repeated escalations of trade friction, technological decoupling, and geopolitical competition are subjecting markets to prolonged "policy noise" and pressure for risk repricing (Rogers et al., 2024).

Gold plays a pivotal role in this process. As a globally recognized safe-haven asset, it provides investors with risk-hedging tools, while its price fluctuations reflect dynamic market expectations of future uncertainties (Burdekin and Tao, 2021). Since gold's value primarily depends on shifts in risk perception and risk premiums rather than discounted cash flows, its market performance serves as a crucial signal for gauging geopolitical tension levels (Feder-Sempach et al., 2024).

Theoretically, geopolitical conflicts constitute "slow-moving risks". Through continuous information accumulation and

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adjustments to expectations, they alter investors' perceptions of discount rates and risk premiums, thereby triggering the "risk repricing effect" in asset prices (Gupta et al., 2024). Within this framework, long-term fluctuations in gold prices can be viewed as the outcome of geopolitical risk transmission and reassessment (Banerjee et al., 2024).

Despite growing research, the literature on geopolitical risk and gold volatility remains incomplete. Quantitative studies on U.S.-China relations, a critical political risk, are limited, and their long-term predictive power and risk pricing effects lack systematic validation. Existing research predominantly focuses on short-term earnings shocks, with insufficient exploration of long-term mechanisms of volatility and heterogeneous responses to "escalation" versus "de-escalation" shocks.

Therefore, this study adopts a risk pricing perspective and introduces the U.S.-China Tension Index (UCT). Using the GJR-GARCH-MIDAS model, it assesses the impact of geopolitical tensions on the long-term volatility of daily log returns in the gold spot market. Results indicate that UCT exerts a significant positive effect on gold's long-term volatility, meaning heightened U.S.-China tensions substantially amplify market volatility. This result remains robust after controlling for monthly realized volatility (RV). Moreover, decomposing UCT into positive shocks (UCT^+) and negative shocks (UCT^-), with UCT^+ exerting a stronger impact than UCT^-, suggests that both escalation and de-escalation drive increases in gold's long-term volatility, which indicates that geopolitical shifts themselves constitute a persistent source of market risk. Out-of-sample forecasting tests also show that models incorporating RV alongside UCT^+ and UCT^- deliver superior long-term volatility predictions.

This study contributes in three key ways: First, it quantifies the impact of U.S.-China relations on global safe-haven asset prices from a bilateral perspective; Second, it uncovers the reverse asymmetric impact of rising versus easing geopolitical tensions on gold's long-term volatility; Third, it reexamines the gold market's long-term response through a risk repricing framework. The findings suggest that gold not only reflects investors' demand for safe havens but also mirrors global risk sentiment and geopolitical cycles, offering significant implications for macroprudential regulation and early warning of systemic risk.

2. Theoretical mechanisms and literature foundation

2.1. Slow-moving nature of geopolitical tensions and their role as systemic risks

Modern asset-pricing theory posits that systematic and non-diversifiable shocks alter investors' expectations regarding future macroeconomic conditions, discount-rate dynamics, and compensation for bearing long-horizon risks, thereby affecting both asset valuations and volatility structures (Bekaert et al., 2014). Geopolitical tensions constitute a distinctive class of such shocks, characterized by gradual information release, structural persistence, and multi-dimensional uncertainty spanning political, economic, and institutional domains. These features align closely with the concept of slow-moving risks introduced by Giglio et al. (2021), which emphasizes that certain shocks unfold over extended horizons and accumulate through gradual adjustments in beliefs rather than abrupt revisions. Geopolitical tensions are reshaping expectations for global trade architecture, supply-chain stability, strategic alliances, and long-run policy direction. Because gold's valuation is predominantly driven by risk compensation rather than cash-flow discounting, these slow-moving risks embed directly into its long-term volatility dynamics, making gold an especially sensitive barometer of persistent geopolitical uncertainty (Banerjee et al., 2024).

2.2. Risk-premium channel: long-horizon transmission of uncertainty shocks

The risk-premium channel provides a primary mechanism through which geopolitical tensions transmit into gold's long-term volatility. Bloom (2009) documents that uncertainty shocks elevate investors' risk aversion and lead to persistent increases in required risk premia. These effects are amplified in environments where uncertainty concerns the long-run macroeconomic and institutional trajectory. Pastor and Veronesi (2012) further show that political uncertainty primarily affects asset prices through variations in discount rates rather than through changes in fundamentals. For an asset such as gold, whose price is not tied to cash flows, the discount-rate channel becomes the dominant source of valuation adjustments (Cheng et al., 2023). Heightened geopolitical uncertainty increases the compensation investors demand for bearing long-term risks, thereby shifting the long-run component of volatility upward (Li et al., 2023). Importantly, this mechanism persists even after controlling for high-frequency realized volatility, underscoring the distinct role of slow-moving geopolitical risk in shaping long-horizon volatility dynamics.

2.3. Information asymmetry and expectation divergence: explaining direction-insensitive volatility

Information-based theories offer an additional pathway linking geopolitical tensions to gold volatility. According to Pástor and Veronesi (2013), greater uncertainty exacerbates information frictions and widens the dispersion of investors' beliefs, resulting in higher volatility even when fundamentals remain unchanged. Geopolitical events are typically accompanied by opaque policy signals, delayed information disclosure, and ambiguous strategic communication, making it difficult for markets to form consensus expectations. Crucially, these uncertainties arise not only during periods of escalation but also during phases of de-escalation, where policy transitions, negotiations, and implementation uncertainties create new information asymmetries. This provides a compelling explanation for the empirically observed phenomenon that both positive and negative shifts in geopolitical tensions raise gold's long-term volatility. Markets react not to the direction of geopolitical shocks per se, but to the degree to which such shocks heighten informational incompleteness and expand belief heterogeneity (Li and Liu, 2025).

2.4. Behavioral mechanisms: amplification through risk perception and ambiguity aversion

Behavioral finance further deepens understanding of why geopolitical tensions persistently affect gold volatility. Under high uncertainty, investors tend to overweight tail risks, exhibit ambiguity aversion, and rely on context-dependent heuristics when evaluating future states of the world (Bordalo et al., 2020). These psychological tendencies induce asymmetric sensitivity to news: escalation triggers immediate flight-to-safety behavior, while de-escalation does not reduce risk perception at the same speed due to inertia in ambiguity attitudes. Thus, even improvements in geopolitical conditions may be interpreted as transitional phases rather than genuine resolution, leaving long-term volatility elevated. This behavioral amplification reinforces the notion that gold's long-term volatility contains a structural component driven by persistent risk perception rather than solely by contemporaneous shocks, explaining the cross-period persistence observed in empirical volatility measures (Gkillas et al., 2020).

2.5. Gold's dual function: safe-haven asset and forward-looking risk-assessment instrument

Gold's role in global financial markets extends beyond its classic function as a safe-haven asset (Baur and Lucey, 2010). Recent literature emphasizes its dual nature as both a defensive asset and a forward-looking instrument for pricing global uncertainty (Burdekin and Tao, 2021; Gupta et al., 2024). As a highly liquid, globally traded, and politically neutral asset, gold absorbs not only immediate flight-to-safety flows but also investors' forward-looking assessments of geopolitical tensions, strategic rivalry, and long-term systemic risk (Baur and Smales, 2020). This dual function means that gold prices embed expectations of future geopolitical cycles, including potential escalation, strategic realignment, and institutional instability. As a result, long-term volatility reflects both realized shocks and anticipatory risk-pricing processes, positioning gold as a key transmitter of slow-moving geopolitical risk into global asset markets (Selmi et al., 2022).

2.6. Structural importance of U.S.–China relations and gaps in existing research

As the most systemically influential major power relationship in the world today, U.S.-China relations profoundly shape international trade patterns, capital flows, technological competition trajectories, and institutional stability (Hanson, 2020). Fluctuations in the intensity of this relationship often drive global risk premiums. However, existing literature exhibits shortcomings in three areas: First, most studies focus on short-term market reactions to geopolitical shocks, lacking systematic examination of how these shocks influence long-term gold volatility as slow variables; Second, there is insufficient decomposition-based identification of directional heterogeneity effects between escalation and de-escalation; Third, there is insufficient recognition of gold's structural role in global risk repricing, particularly how it reflects the cumulative and forward-looking nature of geopolitical risks in long-term volatility. Based on this, this study introduces the UCT and its directional decomposition indicators, employing the GJR-GARCH-MIDAS model to systematically reveal, from a long-term volatility perspective, how Sino-US tensions influence the structural volatility process in the gold market.

3. Methodology

3.1. GJR-GARCH-MIDAS model

This study draws on the research of Conrad and Kleen (2020) and employs the GJR-GARCH-MIDAS model to effectively integrate high-frequency financial data with low-frequency macroeconomic indicators.

$$r_{i,t} = \mu + \sqrt{\tau_t{}^* g_{i,t}}\varepsilon_{i,t} \qquad \forall \mathrm{i} = 1, \ldots, \mathrm{N_t} \tag{1}$$

In Eq. (1), μ denotes the unconditional mean, t represents the monthly low-frequency time dimension, and $r_{i,t}$ is the daily log return of the spot gold price. The model decomposes the conditional variance into two components: $g_{i,t}$, which captures the short-term volatility characteristics of high-frequency data, and τ_t, which reflects the long-term impact of low-frequency macroeconomic variables.

$$g_{i,t} = \left(1 - \alpha - \frac{\gamma}{2} - \beta\right) + \left(\alpha + \gamma^* I_{\{\varepsilon_{i-1,t}<0\}}\right)\frac{\varepsilon_{i-1,t}^2}{\tau_t} + \beta g_{i-1,t} \tag{2}$$

In Eq. (2), the short-term volatility term $g_{i,t}$ is modeled using a GJR-GARCH (1,1) model to characterize the high-frequency volatility characteristics of daily gold returns. The coefficient γ represents the asymmetric effect in the market, reflecting the differential impact of positive and negative shocks on volatility. To ensure the validity and stability of the model, the parameters must satisfy the following conditions: $\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta + \frac{\gamma}{2} < 1$, thereby ensuring the positivity of the conditional variance and cointegration stability.

$$ln\tau_t = m + \theta_X \sum_{k=1}^{K} \varphi_k(w_1, w_2) X_{t-k} \tag{3}$$

In Eq. (3), this study primarily focuses on changes in the long-term coefficient theta to assess the explanatory power of different low-frequency variables for the long-term volatility of the gold market. The long-term volatility term, cap X_{t-k}, is composed of the

realized volatility and the rate of change of the UCT at the Kth lag; its logarithm is taken to ensure that the long-term volatility term, τ_t is always positive. The weighting function $\varphi_k(w_1, w_2)$ is a Beta distribution function, whose parameters w_1 and w_2 control the weight distribution of each lagged variable, used to characterize the dynamic influence path of long-term information on volatility.

$$\varphi_k(w_1, w_2) = \frac{\left(\frac{k}{K}\right)^{w_1-1}\left(1-\frac{k}{K}\right)^{w_2-1}}{\sum_{j=1}^{K}\left(\frac{j}{K}\right)^{w_1-1}\left(1-\frac{j}{K}\right)^{w_2-1}} \tag{4}$$

The Beta weighting function in Eq. (4) is used to characterize the lagged impact of low-frequency variables on high-frequency volatility and its diminishing effect. To maintain the rationality of the model, this study also sets the parameter $w_1 = 1$, ensuring that the impact of exogenous variables on long-term volatility exhibits a monotonic decreasing trend.

The specific form is shown in Formula (5):

$$\varphi_k(w_2) = \frac{\left(1-\frac{k}{K}\right)^{w_2-1}}{\sum_{j=1}^{K}\left(1-\frac{j}{K}\right)^{w_2-1}} \tag{5}$$

Therefore, Eqs. (1), (2), (3), and (5) together constitute the complete form of the GJR-GARCH-MIDAS single-factor model.

This study further extends Eq. (3) to construct a two-factor GJR-GARCH-MIDAS model to simultaneously incorporate two low-frequency variables. The core objective of this setting is to investigate whether the UCT has independent and significant predictive power in explaining the long-term volatility of the gold market after controlling for the market status information represented by realized volatility.

As Chauvet et al. (2015) pointed out, realized volatility typically reflects the market's high sensitivity and immediate response to current macroeconomic conditions. Therefore, if a variable can significantly improve the predictability of volatility after controlling for realized volatility, it indicates that it contains incremental information beyond the current market conditions, demonstrating strong forward-looking and systemic risk disclosure capabilities. The two-factor model was designed precisely to identify and validate whether UCT possesses such independent informational value.

$$ln\tau_t = m + \theta_{RV}\sum_{k=1}^{K}\varphi_k(w_1, w_2)RV_{t-k} + \theta_{UCT}\sum_{k=1}^{K}\varphi_k(w_1, w_2)UCT_{t-k} \tag{6}$$

In Eq. (6), realized volatility is calculated as $RV_{t-k} = \sum_{k=1}^{N_t} r_{t-k}^2$, which is the sum of the squares of the daily log returns of gold spot prices, yielding the realized volatility for the corresponding month. Therefore, Eqs. (1), (2), (5), and (6) constitute the two-factor GJR-GARCH-MIDAS model.

Furthermore, this model can be extended to a multi-factor framework. Following the research in (Li et al., 2025), this study decompose the impact of UCT on long-term volatility in Eq. (6) into positive volatility ($UCT^{+}_{t-k} = \sum_{j=1}^{N} UCT_{t-k} I_{\{UCT_{t-k}>0\}}$) and negative volatility ($UCT^{-}_{t-k} = \sum_{j=1}^{N} UCT_{t-k} I_{\{UCT_{t-k}<0\}}$), which capture the intensification and easing of UCT, respectively. Here, UCT^{+} measures periods when bilateral tensions are rising, while UCT^{-} reflects periods when tensions are declining. This decomposition allows us to investigate potential asymmetries in the effects of geopolitical tensions on long-term volatility.

$$\begin{aligned} ln\tau_t = m + \theta_{RV}\sum_{k=1}^{K}\varphi_k\left(w_{1,RV}, w_{2,RV}\right)RV_{t-k} + \theta^{+}_{UCT}\sum_{k=1}^{K}\varphi_k\left(w^{+}_{1,UCT}, w^{+}_{2,UCT}\right)UCT^{+}_{t-k} \\ +\theta^{-}_{UCT}\sum_{k=1}^{K}\varphi_k\left(w^{-}_{1,UCT}, w^{-}_{2,UCT}\right)UCT^{-}_{t-k}\# \end{aligned} \tag{7}$$

Therefore, Eqs. (1), (2), (5), and (7) constitute the extended setting of the GJR-GARCH-MIDAS Multi-factor model.

All model parameters are estimated using Gaussian quasi-maximum likelihood (QMLE). While the likelihood function is specified under the conditional normality assumption for estimation purposes, statistical inference does not rely on this assumption. Instead, standard errors are computed using the outer-product-of-gradients (OPG) method, which is robust to deviations from normality and heavy-tailed innovations commonly observed in financial time series (Wang and Ghysels, 2015). Accordingly, all reported statistical inferences for the MIDAS coefficients and other parameters are based on these robust standard errors.

3.2. Model evaluation methods

This study employs three commonly used methods to evaluate the reliability of models' out-of-sample predictions. First, the Model Confidence Set (MCS) method is utilized to systematically compare the out-of-sample forecasting performance across models, thereby identifying statistically superior forecasting approaches (Hansen et al., 2011). Second, the DM test proposed by Diebold and Mariano (2002) and the out-of-sample R^2 metric introduced by Rapach et al. (2010) are employed to examine whether the models demonstrate a significant advantage over the benchmark model in reducing prediction errors.

3.2.1. Model confidence set (MCS) test

First, this study employs the Model Confidence Set (MCS) testing method proposed by Hansen et al. (2011). This method, based on the prediction loss function, progressively eliminates models with significantly poorer performance at a given confidence level ($p = 0.10$), ultimately forming a set of models with relatively superior predictive capabilities. Furthermore, relying solely on a single loss function proves insufficient for accurately assessing the precision of volatility forecasts (Diebold and Lopez, 1996). To address this limitation, this paper employs three loss function metrics to comprehensively evaluate model prediction errors: Mean Squared Error (MSE), Mean Absolute Error (MAE), and Log R-squared (R^2LOG).

The formulas are as follows:

$$MSE = \frac{1}{N}\sum_{n=1}^{N}\left(\sigma_t^2 - \widehat{\sigma}_t^2\right)^2 \# \tag{8}$$

$$MAE = \frac{1}{N}\sum_{n=1}^{N}\left|\sigma_t^2 - \widehat{\sigma}_t^2\right| \# \tag{9}$$

$$R^2LOG = \frac{1}{N}\sum_{n=1}^{N}\left(ln\frac{\sigma_t^2}{\widehat{\sigma}_t^2}\right)^2 \# \tag{10}$$

Among these, σ_t^2 is a proxy for the actual volatility (the square of the daily logarithmic returns of gold); $\widehat{\sigma}_t^2$ denotes the volatility estimate predicted by the model.

3.2.2. Diebold and Mariano (DM) test

Additionally, this study employs the Diebold and Mariano (2002) diagnostic test for comparing forecasting accuracy (DM test) to evaluate the predictive performance of each model relative to the benchmark GJR-GARCH(1,1) model. The null hypothesis of the DM test posits that the two forecasting models possess statistically equivalent predictive capabilities. When the test yields a p-value below the preset significance level, the null hypothesis is rejected, indicating a significant difference in forecasting performance between the two models.

The statistical quantity of the DM test is calculated as follows:

$$DM = \frac{\overline{d}}{\sqrt{var(d_t)}} \sim N(0,1) \# \tag{11}$$

Among them,

$$d_s = Loss_{i,t}^2 - Loss_{benchmark,t}^2 \# \tag{12}$$

$Loss_{i,t}$ and $Loss_{benchmark,t}$ represent the prediction errors of models i and benchmark model at time t, respectively. $\overline{d}$ is the sample mean of the error difference sequence d_t, and $var(d_t)$ is its variance estimate. The null hypothesis for the DM test is that model i does not differ from the baseline model in terms of predictive accuracy. Therefore, if the DM statistic is significantly negative, it indicates that model i has higher predictive accuracy than the baseline model.

Furthermore, this study measures the improvement in predictive loss of the enhanced model relative to the baseline model by calculating the loss ratio between the two models. Specifically, let α_1 denote the loss function value of the extended GJR-GARCH-MIDAS model and α_2 denote that of the traditional GJR-GARCH(1,1) model. The loss ratio between the two models is then expressed as α_1/α_2. When this ratio is less than 1, and the corresponding DM test statistic is significantly negative, it indicates that the extended model is statistically significantly superior to the baseline model.

3.2.3. Out-of-sample R^2_{oos} test

Finally, this study employs the out-of-sample R^2 (R^2_{oos}) test proposed by Rapach et al. (2010) to reassess the predictive capability of GJR-GARCH-MIDAS-type models relative to the benchmark GJR-GARCH(1,1) model. The null hypothesis states that the mean squared error (MSE) of the benchmark model is no greater than the MSE of predictive model i. Therefore, when a model's R^2_{oos} value is positive and statistically significant, the null hypothesis is rejected. This indicates that the model's prediction error is significantly lower than the benchmark model, demonstrating superior out-of-sample forecasting performance (Clark and West, 2007).

The formula for out-of-sample R^2 is as follows:

$$R_{oos}^2 = 1 - \frac{\sum_{t=1}^{N}\left(\sigma_t^2 - \widehat{\sigma}_{i,t}^2\right)^2}{\sum_{t=1}^{N}\left(\sigma_t^2 - \widehat{\sigma}_{benchmark,t}^2\right)^2} \# \tag{13}$$

Among these, σ_t^2 serves as a proxy for the true volatility (the square of the logarithmic returns of the spot gold price); $\widehat{\sigma}_{i,t}^2$ denotes the forecasted volatility from predictive model i; while $\widehat{\sigma}_{benchmark,t}^2$ represents the forecasted volatility from the benchmark model GJR-GARCH(1,1).

4. Data sources

4.1. Data description

The daily spot gold prices used in this study are sourced from the Investing.com database, with market volatility measured by daily log returns. The RV indicator is calculated by summing the squared daily log returns over the month. The UCT index is derived from Rogers et al. (2024), in which the first-order log difference is taken and multiplied by 100 to reflect its monthly rate of change. Due to data availability constraints, the sample period is set from January 4, 1993, to February 29, 2024.

As shown in Table 1, the Jarque–Bera test rejects the null of normality, and the realized gold returns exhibit pronounced skewness and excess kurtosis, confirming heavy-tailed and non-normal behavior. Therefore, statistical inference based on OPG robust standard errors is appropriate, ensuring reliable conclusions despite deviations from normality. Moreover, both ADF and PP tests confirm that the three series are stationary. The ARCH test further indicates significant conditional heteroskedasticity in the residuals, suggesting that GARCH-type models are suitable for capturing their volatility structure.

Fig. 1 illustrates the temporal evolution of gold returns alongside the decomposed UCT^{+} and UCT^{-} components. During multiple significant geopolitical and financial shocks, all three components exhibit synchronized fluctuations, revealing a pronounced resonance relationship between shifts in geopolitical uncertainty and volatility in the gold market.

4.2. Identification and exogeneity of the UCT

A key identifying assumption in this study is that the UCT is exogenous to short-term fluctuations in gold market volatility. In theory, financial market volatility may influence media narratives or market sentiment, which, in turn, could affect indicators of geopolitical tensions. However, the UCT is constructed based on large-scale textual analysis, capturing structural interactions between the United States and China at political, economic, and strategic levels. Its evolution is primarily driven by policy decisions and geopolitical events rather than short-term financial market fluctuations. Therefore, at a monthly frequency, it is unlikely that gold market volatility systematically affects UCT.

To further examine this assumption, we conduct Granger causality tests using a vector autoregression (VAR) model to assess the dynamic relationship between gold market volatility and UCT. The lag length of the VAR model is determined according to the Akaike Information Criterion (AIC). AIC is chosen because it effectively balances model fit and complexity, especially for samples of limited size, and tends to select slightly longer lags than other criteria, such as BIC. This reduces the risk of omitted variable bias while better capturing the dynamic interactions between variables (Ding et al., 2018). Table 2 presents the test results. The results indicate that gold market volatility does not Granger-cause UCT, as the null hypothesis cannot be rejected at conventional significance levels. In contrast, UCT significantly predicts gold market volatility, with the null hypothesis of "UCT does not Granger-cause gold volatility" rejected at the 10% significance level.

These findings suggest that gold market volatility does not have a statistically significant impact on UCT, alleviating potential concerns regarding reverse causality. This provides support for treating UCT as an important exogenous driver of gold market volatility and reinforces the robustness of the empirical identification strategy adopted in this study.

5. Empirical results

5.1. In-sample estimation results analysis

The in-sample parameter estimation results for each model (Table 3) indicate that parameters characterizing short-term volatility satisfy the constraints ($\alpha > 0$, $\beta \geq 0$, and $\alpha + \beta + \frac{\gamma}{2} < 1$) and are statistically significant at the 1% level. This confirms the presence of

Table 1
Descriptive statistics.

Variable	XAU_USD	RV	UCT
Mean	0.0224	21.1162	0.3764
Std.Dev	0.9865	24.2332	18.8108
Min	−8.8756	0.7729	−93.9705
Max	10.4371	195.9086	87.1417
J-B	21,443.0767***	6040.6927***	108.8943***
ADF test	−34.9022***	−5.3479***	−9.6558***
PP test	−90.7873***	−11.2095***	−42.9495***
ARCH(10)	441.8117***	118.0940***	29.5621***
Obs	8113	374	374

Note: J-B stands for the Jarque-Bera test for normal distribution, whose null hypothesis is that the series follows a normal distribution. ADF stands for the Augmented Dickey-Fuller unit root test, and PP denotes the Phillips-Perron unit root test; for both ADF and PP tests, the null hypothesis is that the series contains a unit root, i.e., is non-stationary. ARCH(10) represents Engle's LM test for conditional heteroskedasticity with 10 lags, where the null hypothesis is no ARCH effects. ***, **, * denote rejection of the null hypothesis at the 1%, 5%, and 10% significance levels, respectively.

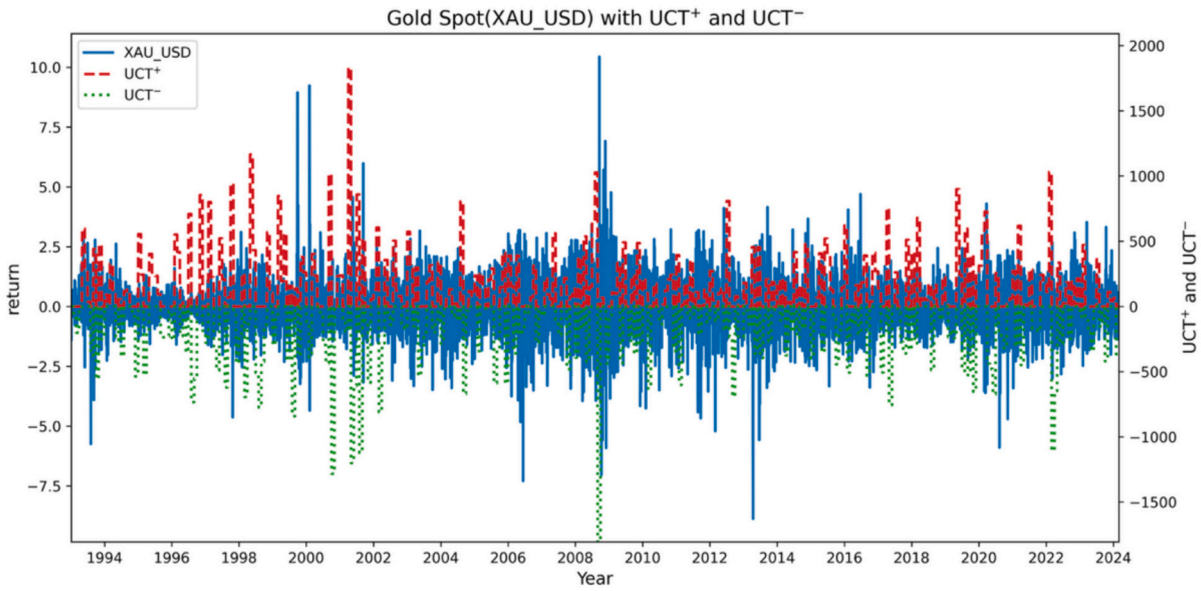

Fig. 1. Gold Spot Price Movements and the US-China Tension Index. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Granger causality test.

Equation	Excluded	Chi2	df	Prob>Chi2
RV	UCT	11.571	6	0.072*
UCT	RV	7.178	6	0.305

Note: The tests are conducted using a VAR model, with the optimal lag length selected based on the Akaike Information Criterion (AIC). The null hypothesis is that the excluded variable does not Granger-cause the dependent variable. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

pronounced volatility clustering in the spot gold market. Notably, the asymmetry parameter γ is significantly negative across all models, revealing a classic "anti-leverage effect" in the gold market (Todorova, 2017). This implies that positive shocks amplify volatility more effectively than negative shocks. During price increases, market trading activity and capital inflows often intensify simultaneously, thereby magnifying short-term volatility reactions (Chang et al., 2021).

Further examination of the estimation results for the long-term volatility term θ reveals that the realized volatility coefficient $\theta_{RV}$is significantly positive, indicating that monthly realized volatility exerts a persistent positive influence on the long-term volatility of gold returns, which is consistent with the mechanism whereby short-term market stress spills over into the long-term risk structure. Concurrently, the UCT coefficient θ_{UCT} is also significantly positive, indicating that heightened U.S.-China geopolitical tensions substantially elevate the long-term volatility of the gold market (Wu, 2025). When policy conflicts intensify, market concerns about future uncertainties are amplified, thereby significantly elevating the long-term variance of gold returns. Even after controlling for realized volatility, the θ_{UCT} remains significant and positive, indicating that the degree of U.S.-China tensions exerts an independent and enduring influence on long-term volatility in the gold market, rather than merely being a secondary effect of short-term market fluctuations.

When the UCT is further decomposed into UCT^{+} and UCT^{-}, both θ coefficients remain significantly positive, indicating that geopolitical shifts amplify gold market volatility regardless of whether U.S.–China relations are escalating or de-escalating. However, the magnitude of their effects differs markedly: the estimated coefficient for UCT^{+} (0.012) is substantially larger than that for UCT^{-} (0.004), and a Wald test strongly rejects the null hypothesis of equal coefficients (Wald statistic = 23.9782, p-value = 9.74×10^{-7}). This evidence demonstrates a pronounced reverse asymmetric effect in the gold market's response to geopolitical tensions: upward shifts (tension escalation) amplify volatility more strongly than downward shifts (tension easing) (Chang et al., 2021). On one hand, escalating tensions heighten risk aversion, prompting safe-haven capital inflows (Chiarella et al., 2016). On the other hand, even during periods of relative easing, the persistence of geopolitical uncertainty and the potential for renewed tensions lead market participants to maintain cautious expectations regarding the sustainability of de-escalation measures. Consequently, gold trading activity remains elevated, and long-term volatility does not decline immediately.

Table 3
In-sample estimates of model parameters.

	Model 1	Model 2	Model 3	Model 4
μ	0.010	0.008	0.010	0.010
	(0.010)	(0.010)	(0.010)	(0.010)
α	0.059***	0.054***	0.057***	0.058***
	(0.002)	(0.002)	(0.002)	(0.003)
β	0.947***	0.955***	0.948***	0.946***
	(0.001)	(0.001)	(0.001)	(0.002)
γ	−0.026***	−0.024***	−0.026***	−0.023***
	(0.003)	(0.002)	(0.003)	(0.003)
m	−0.473***	0.069	−0.604***	−1.906***
	(0.137)	(0.148)	(0.140)	(0.222)
θ_{RV}	0.027***		0.030***	0.030***
	(0.005)		(0.005)	(0.005)
θ_{UCT}		0.141***	0.157***	
		(0.020)	(0.021)	
θ_{UCT^+}				0.012***
				(0.001)
θ_{UCT^-}				0.004***
				(0.001)
$w_{2,\mathrm{RV}}$	1.110***		1.000***	1.000***
	(0.317)		(0.244)	(0.256)
$w_{2,\mathrm{UCT}}$		1.000***	1.000***	
		(0.063)	(0.059)	
w_{2,UCT^+}				1.000***
				(0.058)
w_{2,UCT^-}				1.000***
				(0.178)
LLH	−9781.536	−9780.370	−9770.918	−9761.023
AIC	19,577.073	19,574.739	19,559.837	19,544.046
BIC	19,625.374	19,623.040	19,621.938	19,619.947

Note: This table reports the parameter estimates of the factor models specified with varying lag structures. LLH is the log-likelihood function, AIC denotes the Akaike information criterion, and BIC indicates the Bayesian information criterion. The numbers in parentheses are OPG-based robust standard errors. ***, **, and * denote respectively the significance levels at 1%, 5%, and 10%.

5.2. Out-of-sample forecast analysis

Compared with in-sample fit accuracy, market participants are more concerned with a model's predictive capability for future market volatility than with its ability to explain historical data. To enhance the robustness and reliability of out-of-sample forecasts, this study employs a combined assessment using three widely accepted academic validation methods: the Model Confidence Set (MCS), the Diebold–Mariano (DM) test, and the out-of-sample coefficient of determination (R^2_{oos}) test.

This study employs a rolling time window approach to systematically evaluate the model's out-of-sample predictive capability under varying market conditions. Specifically, the model is estimated using a fixed-length in-sample data set for each period, followed by forecasting the subsequent out-of-sample period. After completing the forecast, the estimation window is shifted forward by one period, repeating this process until the entire out-of-sample period is covered. To this end, the data are divided into two subsets: in-sample data (January 4, 1993, to December 31, 2013) for model parameter estimation, and out-of-sample data (January 1, 2014, to February 29, 2024) for model forecasting performance evaluation.

The out-of-sample period covers both relatively stable phases (2014–2018) and periods of significantly heightened volatility (2019–2024) in the gold market. The former period coincided with a phase of moderate global economic recovery, characterized by weak market risk aversion and limited gold price fluctuations. The latter period, however, was marked by persistent shocks from major geopolitical and macroeconomic uncertainties, including the COVID-19 pandemic, escalating trade frictions, and semiconductor

Table 4
MCS test results.

2646 days	*MSE*		*MAE*		R^2LOG	
	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
$GJR-GARCH(1,1)$	0.486	0.326	0.001	0.004	0.001	0.000
Model 1	0.486	0.274	0.002	0.089	0.004	0.015
Model 2	0.486	0.562	0.001	0.089	0.001	0.015
Model 3	0.486	0.562	0.010	0.089	0.010	0.015
Model 4	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Note: The numbers in the table are *p*-values. Bold and underlined values indicate the model achieving the highest predictive accuracy in each comparison. The time horizon is 2646 days.

restrictions, leading to substantially heightened market volatility. Incorporating both market states within a unified forecast period enables comprehensive testing of the model's predictive performance under both normal and high-uncertainty scenarios, thereby providing a more reliable assessment of its robustness and applicability.

5.2.1. Model confidence set (MCS) test

Based on the MCS validation results (Table 4), model performance differs significantly across the three distinct loss-function settings. Under Mean Squared Error (MSE), all models passed validation, with Model 4, which simultaneously incorporates RV, UCT^{+}, and UCT^{-}, demonstrating optimal performance (p-value = 1) and markedly outperforming the other models. Model 4 also maintained its leading position under the Mean Absolute Error (MAE) and Logarithmic R-squared (R^2LOG) loss functions. Overall, the MCS test results indicate that Model 4 demonstrates robust out-of-sample predictive performance across different loss-function settings.

5.2.2. Diebold and Mariano (DM) test

To examine differences in predictive performance among models, this study employs the DM test proposed by Diebold and Mariano (2002), using the GJR-GARCH(1,1) model as the benchmark for comparison. Results (Table 5) show that, under the MSE criterion, all models exhibit lower prediction losses than the benchmark model, with Model 4 performing the best. Although differences did not reach statistical significance, overall prediction errors were small. Under MAE and R^2LOG, all GJR-GARCH-MIDAS models significantly outperformed the benchmark model. Model 4 reduced prediction loss by approximately 1.0%, 1.9%, and 1.9%, respectively, compared to the benchmark, demonstrating its relative advantage in out-of-sample predictions.

5.2.3. Out-of-sample R^2_{oos} test

Finally, to further evaluate the predictive effectiveness of the models, this study employs R^2_{oos} tests to compare each model. Empirical results (Table 6) indicate that, except for Model 1, the R^2_{oos} values of all other models are significantly positive, indicating that these models exhibit superior predictive fit compared to baseline models in out-of-sample predictions. Among these, Model 4 achieves the highest R^2_{oos} of 0.99%. While the magnitude of this gain appears modest in economic terms, it is statistically significant and represents a meaningful improvement given the inherent difficulty in forecasting asset volatility. This result further validates that, within the benchmark GJR-GARCH(1,1) framework, the composite model that simultaneously incorporates RV, UCT^{+}, and UCT^{-} achieves optimal performance.

In summary, the results from the three out-of-sample prediction evaluation methods show high consistency, all indicating that Model 4 exhibits optimal predictive performance across different loss functions and statistical tests. This outcome highlights the complementary role of RV and UCT decomposition metrics in volatility forecasting, validating the effectiveness of multi-source information fusion in enhancing prediction accuracy.

6. Robustness checks

Given that out-of-sample forecasting performance may vary with the selection of the forecast period, this paper further adjusts the forecast intervals and conducts segmented evaluations to ensure robust results. Building upon the original settings, the out-of-sample period is divided into two distinct market phases: a low-volatility phase (January 1, 2014–December 31, 2018, 1300 trading days) and the high-volatility phase (January 1, 2019–February 29, 2024, 1345 trading days). This focuses on testing the model's predictive performance under both normal market conditions and high-uncertainty environments.

The former corresponds to a period of relatively stable global economic conditions and low overall financial market volatility, during which gold price movements were more continuous and predictable. The latter encompasses multiple major uncertainties, including the COVID-19 pandemic, escalating trade frictions, supply chain disruptions, and semiconductor controls, which subject markets to intense exogenous shocks and significantly alter volatility structures.

Conducting separate predictive tests under these conditions helps evaluate the model's adaptability and robustness in extreme

Table 5
Results of the Diebold–Mariano test between GJR-GARCH(1,1) and each models.

DM-2646 days	*MSE*	*MAE*	R^2LOG
Model 1	0.999 (−0.659)	0.996*** (−4.471)	0.995*** (−7.569)
Model 2	0.993* (−1.846)	0.994*** (−3.185)	0.997*** (−3.406)
Model 3	0.992 (−1.556)	0.990*** (−4.823)	0.991*** (−8.063)
Model 4	**0.990** **(−1.731)**	**0.981***** **(−8.272)**	**0.981***** **(−13.476)**

Note: This table displays the results of Diebold-Mariano forecast evaluation for various models relative to the GJR-GARCH(1,1) model as the benchmark. Ratios less than 1 indicate that the model's forecast for volatility is superior to the benchmark model. DM test statistics are shown in parentheses. The time horizon is 2646 days. ***, **, * represent rejecting the null hypothesis at the 1%, 5%, and 10% significance levels, respectively.

Table 6
R^2_{oos} test.

2646 days	$R^2_{oos}(\%)$	$MSPE-adjusted$	$p-value$
Model 1	0.080	0.004	0.132
Model 2	0.690***	0.025	0.008
Model 3	0.800**	0.031	0.015
Model 4	0.990***	0.038	0.008

Note: This table reports the out-of-sample forecasting performance based on the out-of-sample R^2 test proposed by Clark and West (2007). The null hypothesis states that the benchmark model has significantly smaller or equal mean squared error (MSE) compared to the evaluated model i. In this study, the benchmark model is specified as the GJR-GARCH(1,1) model. A positive value of out-of-sample R^2 indicates that the forecasting model of interest outperforms the benchmark model in terms of predictive accuracy. ***, **, and * denote respectively the significance levels at 1%, 5%, and 10%. The time horizon is 2646 days.

uncertainty scenarios. By independently forecasting and comparing the two subsamples, this study identifies whether the model's predictive advantages remain consistent across different market states, thereby providing a more comprehensive validation of its out-of-sample effectiveness.

6.1. Model confidence set (MCS) test

The MCS test results across two distinct market cycles (Table 7) demonstrate that Model 4, incorporating RV alongside UCT^+ and UCT^-, exhibits significant predictive advantages regardless of whether the market is in a low- or high-volatility phase. During low-volatility periods (Panel A), Model 4 remains within the optimal model set under all three loss functions, reflecting its superior ability to capture volatility structures in low-uncertainty environments. During high-volatility periods (Panel B), despite exogenous shocks such as the pandemic, trade frictions, and supply chain disruptions that led to more pronounced nonlinear volatility, Model 4 maintained optimal performance under both MSE and R^2LOG loss functions, demonstrating its robustness and resilience to disturbances amid severe market turbulence. Overall, Model 4 consistently ranks among the optimal models across both market scenarios, aligning with out-of-sample results from mixed MCS samples. This further validates that incorporating both RV and decomposed policy uncertainty indicators into the model effectively enhances the comprehensiveness and cross-cycle applicability of volatility forecasting.

6.2. Diebold and Mariano (DM) test

The results of the DM test across two distinct periods (Table 8) reveal that during the low-volatility period (Panel A), only Model 4, which simultaneously incorporates RV alongside UCT^+ and UCT^-, achieves a prediction loss ratio below 1 across all three loss functions. This indicates superior predictive accuracy relative to the benchmark model GJR-GARCH(1,1). In contrast, during the high-volatility period (Panel B), all models exhibit lower predictive losses than the benchmark. However, Model 4 remains optimal under both MSE and R^2LOG loss functions, demonstrating robustness in environments of extreme uncertainty.

Further examination of loss ratios reveals that Model 4 reduced predictive losses by 0.4%, 1.7%, and 1.7% respectively, across the three loss functions during the low-volatility phase. During high volatility, its prediction losses decreased by 1.6% and 1.8% under MSE

Table 7
MCS test results.

	MSE		MAE		R^2LOG	
	T_R	T_{max}	T_R	T_{max}	T_R	T_{max}
Panel A: Low-volatility-1300 days						
$GJR-GARCH(1,1)$	0.310	0.367	0.009	0.105	0.006	0.063
Model 1	0.310	0.367	0.009	0.105	0.010	0.063
Model 2	0.251	0.367	0.002	0.105	0.000	0.063
Model 3	0.285	0.367	0.005	0.105	0.006	0.063
Model 4	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
Panel B: High-volatility-1345 days						
$GJR-GARCH(1,1)$	0.442	0.470	0.001	0.000	0.000	0.000
Model 1	0.636	0.477	0.150	0.313	0.058	0.037
Model 2	0.262	0.470	0.000	0.313	0.000	0.037
Model 3	0.864	0.864	**1.000**	**1.000**	0.913	0.913
Model 4	**1.000**	**1.000**	0.811	0.811	**1.000**	**1.000**

Note: The numbers in the table are p-values. Bold and underlined values indicate the model achieving the highest predictive accuracy in each comparison.

Table 8
Results of the Diebold–Mariano test between GJR-GARCH(1,1) and each mode.

	MSE	*MAE*	R^2LOG
Panel A: Low-volatility-1300 days			
Model 1	1.001	1.002	0.999
	(0.252)	(0.956)	(−0.090)
Model 2	1.003	1.005	1.003
	(1.081)	(2.263)	(2.399)
Model 3	1.003	1.005	1.003
	(0.870)	(1.868)	(1.985)
Model 4	**0.996**	**0.983*** **	**0.983*** **
	(−1.417)	**(−7.022)**	**(−9.200)**
Panel B: High-volatility-1345 days			
Model 1	0.989	0.993**	0.991***
	(−0.948)	(−2.447)	(−8.254)
Model 2	0.990	0.991***	0.992***
	(−1.621)	(−3.327)	(−4.977)
Model 3	0.985	0.981***	0.983***
	(−1.768)	(−5.308)	(−9.566)
Model 4	**0.984**	0.982***	**0.982*** **
	(−1.480)	(−4.554)	**(−9.142)**

Note: This table displays the results of Diebold-Mariano forecast evaluation for various models relative to the GJR-GARCH(1,1) model as the benchmark. Ratios less than 1 indicate that the model's forecast for volatility is superior to the benchmark model. DM test statistics are shown in parentheses. ***, **, * represent rejecting the null hypothesis at the 1%, 5%, and 10% significance levels, respectively.

and R^2LOG, respectively. Overall, the DM test results across both periods align with out-of-sample evaluations on mixed samples, further validating Model 4's robust predictive capability and overall superiority across market states.

6.3. *Out-of-sample* R^2_{oos} *test*

The out-of-sample coefficient of determination (R^2_{oos}) results for the two periods (Table 9) further corroborate the comparative performance of the models. In the low-volatility period (Panel A), only Model 4 yields a significantly positive R^2_{oos}(0.43%), clearly outperforming all alternative specifications. This indicates that augmenting the baseline GJR-GARCH(1,1) framework with RV and the decomposed UCT^+ and UCT^- measures substantially enhances the model's predictive ability for gold market volatility. In the high-volatility period (Panel B), all models except Model 1 exhibit significantly positive R^2_{oos} values; however, Model 4 again achieves the highest value (1.58%). This consistent positive performance, albeit small in magnitude, confirms the model's robustness under pronounced market uncertainty. Overall, the R^2_{oos} evidence across both periods aligns closely with the findings from the mixed-sample out-of-sample evaluation, reinforcing the conclusion that Model 4 consistently delivers robust and dominant predictive performance across different market conditions.

Table 9
Out-of-sample R^2_{oos} test.

	R^2_{oos}(%)	*MSPE − adjusted*	*p − value*
Panel A: Low-volatility-1300 days			
Model 1	−0.070	0.001	0.438
Model 2	−0.280	−0.002	0.634
Model 3	−0.290	−0.001	0.534
Model 4	0.430**	0.013	0.012
Panel B: High-volatility-1345 days			
Model 1	1.080	0.048	0.123
Model 2	1.010**	0.045	0.022
Model 3	1.520**	0.069	0.012
Model 4	1.580**	0.074	0.027

Note: This table reports the out-of-sample forecasting performance based on the out-of-sample R^2 test proposed by Clark and West (2007). The null hypothesis states that the benchmark model has significantly smaller or equal mean squared error (MSE) compared to the evaluated model i. In this study, the benchmark model is specified as the GJR-GARCH(1,1) model. A positive value of out-of-sample R^2 indicates that the forecasting model of interest outperforms the benchmark model in terms of predictive accuracy. ***, **, and * denote respectively the significance levels at 1%, 5%, and 10%.

Overall, the three out-of-sample forecasting evaluations across both market phases yielded highly consistent conclusions, fully validating the robustness of the research. Regardless of the loss function or statistical test employed, Model 4 consistently demonstrated optimal forecasting performance, indicating its outstanding adaptability and reliability under diverse market conditions. This result further demonstrates that incorporating RV alongside decomposed $UCT^{+}$and UCT^{-} into the volatility modeling framework significantly enhances the accuracy and robustness of long-term volatility forecasting in the gold market. This approach proves effective not only during periods of moderate volatility but also maintains exceptional performance under high uncertainty and strong exogenous shocks, highlighting the model's practical applicability across diverse market conditions.

7. Conclusions and implications

7.1. Conclusions

This study employs the GJR-GARCH-MIDAS model and incorporates the UCT to systematically analyze the impact of geopolitical tensions on long-term volatility in the spot gold market. Key findings are as follows: (1) Gold market investors exhibit a dual behavior of “chasing gains and seeking safety” alongside speculation, amplifying volatility responses to positive shocks; (2) The UCT exerts a significant positive effect on long-term gold volatility, with heightened U.S.-China tensions markedly intensifying market fluctuations; (3) This effect remains robust after controlling for RV, indicating that the UCT contains incremental information beyond traditional volatility metrics; (4) Decomposing UCT into UCT^{+} and UCT^{-} reveals that both escalation and de-escalation amplify long-term gold volatility, with UCT^{+} exerting a notably stronger effect than UCT^{-}, indicating that geopolitical shifts themselves constitute a persistent risk source; (5) Out-of-sample forecasting tests further show that models incorporating both RV and UCT^{+}/UCT^{-} deliver optimal performance in predicting long-term gold volatility.

7.2. Discussions

The empirical findings of this study demonstrate that during periods of heightened geopolitical risk, gold exhibits not only its traditional role as a “safe-haven asset” but also pronounced risk-repricing characteristics (Banerjee et al., 2024). As external uncertainty arising from geopolitical tensions intensifies, investors' expectations regarding future macroeconomic conditions and policy trajectories undergo systematic adjustment (Pástor and Veronesi, 2013). This shift increases demand for long-horizon risk compensation, thereby increasing gold's systematic risk exposure and strengthening its long-term volatility structure (Li et al., 2023). In this sense, gold functions not merely as a passive hedge but as an asset that actively absorbs and reflects macro-level uncertainty through the risk-premium channel (Selmi et al., 2022).

From a theoretical standpoint, the UCT operates as a prototypical slow-moving risk. By influencing multi-period expectations, discount rates, and risk preferences, UCT gradually transmits into the long-run component of asset prices. This mechanism is consistent with Bloom's (2009) argument that uncertainty shocks exert persistent amplification effects, and aligns with Pastor and Veronesi (2012) view that political risks shape asset pricing through the risk-premium channel.

Notably, this study finds that both positive shifts (escalation) and negative shifts (temporary de-escalation) in geopolitical tensions significantly amplify long-term volatility in the gold market, indicating that changes in the geopolitical landscape itself constitute a source of long-term uncertainty. However, the impact intensity of these two types of shocks differs significantly. Compared to de-escalation phases, the volatility amplification effect triggered by escalation demonstrates a markedly stronger reverse asymmetry. This result implies that the gold market is not entirely insensitive to the direction of geopolitical shocks but rather exhibits a more pronounced, persistent risk-repricing response to escalation.

Economically, this asymmetric feature reflects investors' tendency to systematically increase their assessment of future uncertainty when confronted with heightened geopolitical tensions, thereby continuously adjusting their risk-aversion and asset allocation strategies. During periods of relative easing, however, the market's risk-correction process remains relatively slow due to the persistent risk of recurrence and policy uncertainty in geopolitical dynamics. Consequently, gold's safe-haven and store-of-value functions remain elevated, making it difficult for long-term volatility to subside rapidly.

Taken together, these results reveal a more complex and persistent risk-pricing logic underlying gold's behavior in high-uncertainty environments. They suggest that gold's volatility dynamics reflect not only short-term flight-to-safety behavior but also the sustained accumulation of slow-moving geopolitical uncertainty in the long-run component of volatility (Li et al., 2023). This study therefore provides new empirical insights into how geopolitical tensions reshape asset-pricing mechanisms and enriches the understanding of gold's role in periods of elevated global uncertainty.

7.3. Implications

Building on the empirical findings, several policy-relevant implications emerge for financial regulators and market participants. The evidence indicates that geopolitical shocks generate pronounced cross-market transmission effects that extend well beyond the geographical locus of the underlying events. This highlights the need for regulators to strengthen integrated monitoring frameworks that jointly track price dynamics and capital flows across equity, foreign exchange, and commodity markets. During periods of rising risk aversion, gold often serves as a key conduit for rapid cross-market capital flows. Without a systematic mechanism to monitor shifts in its volatility structure and associated flows, localized disturbances may propagate across markets, amplifying structural price distortions and increasing the broader financial system's vulnerability.

From a macro-policy perspective, the long-horizon volatility observed in the gold market reveals the cumulative effects of slowly evolving geopolitical risks, offering policymakers valuable forward-looking information. As a persistent indicator of geopolitical uncertainty, the UCT enhances the informational content of existing systemic risk assessment tools used by central banks. Incorporating such indicators into frameworks for reserve management, capital-flow surveillance, and exchange-rate policy design may strengthen policymakers' ability to identify emerging vulnerabilities. In an environment of sustained geopolitical tension, a moderate increase in the share of gold within official reserves may enhance resilience against external shocks. Nevertheless, any adjustment in reserve composition should be carefully calibrated to a country's risk-absorption capacity, macroeconomic objectives, and external constraints to avoid compromising the flexibility of reserve deployment.

For institutional investors, the strong predictive power of the UCT underscores that geopolitical risk is not merely a short-lived event shock but a quantifiable and forecastable long-term risk factor. Integrating geopolitical risk indicators into strategic asset allocation, risk budgeting, stress testing, and VaR frameworks can improve the robustness of risk measurement and enhance the efficiency of allocating safe-haven assets. For globally diversified portfolios, dynamically adjusting gold holdings and volatility forecasting models in response to changes in geopolitical uncertainty can help stabilize risk-return profiles, particularly during periods of elevated uncertainty.

At the global financial-stability level, fluctuations in U.S.–China geopolitical tensions have the potential to reshape international capital flow patterns and magnify external pressures on emerging markets, including depreciation risks and capital outflows. Against this backdrop, national regulators should enhance cross-border macroprudential coordination by strengthening the shock-absorption capacity of foreign-exchange reserves, improving countercyclical capital buffers, and promoting the exchange of key risk indicators and early warning information. Developing a more resilient and coordinated international regulatory framework can help mitigate the spillover effects of geopolitical shocks and contribute to greater stability within the global financial system.

7.4. Shortcomings and perspectives

Although this study provides new empirical evidence revealing the mechanism by which geopolitical tensions influence long-term volatility in the gold market, several avenues for further expansion remain. First, while the UCT serves as a core representation of geopolitical uncertainty, effectively reflecting shifts in one of the world's most critical bilateral relationships, geopolitical risks themselves exhibit multidimensional, cross-regional, and context-heterogeneous characteristics. Future research could explore incorporating additional geopolitical indicators from diverse sources or integrating narrative text analysis, media event data, and machine learning feature extraction to construct broader, more granular frameworks for measuring geopolitical risk, thereby enriching the dimensions of risk characterization.

Second, this study finds that both UCT^{+} and UCT^{-} significantly amplify long-term gold volatility. This “direction-insensitive” symmetric effect holds significant theoretical implications, though its underlying mechanisms warrant further exploration in subsequent research. This phenomenon may stem from gold's high sensitivity to overall uncertainty, or from the narrative structure of geopolitical events and information asymmetry during policy transition periods. Future research could employ sentiment polarity analysis and narrative theme decomposition to investigate whether different types and contexts of geopolitical shocks exhibit more nuanced directional differences, thereby enhancing our understanding of the mechanisms underlying the formation of long-term volatility.

Finally, this study focuses exclusively on the spot gold market and does not incorporate other potentially affected markets into the analysis, such as silver and platinum markets, gold ETFs and futures markets, or the cryptocurrency market, which has increasingly exhibited safe-haven characteristics in recent years. Different assets may exhibit distinct transmission pathways and risk-pricing logic when facing geopolitical shocks. Future research that conducts cross-sectional comparisons across multiple assets and markets would enhance the external validity of findings and reveal additional mechanisms of systematic risk transmission.

Authors contribution

Y. Liu: Data curation, Methodology, Writing – original draft, Writing – review & editing; S. Li: Data curation, Validation, Writing – original draft, Writing – review & editing. All authors have made significant contributions to the research and agree to be fully accountable for the work presented in this manuscript.

AI software usage

No artificial intelligence (AI) software was used to prepare the manuscript, either as a whole or as a part.

Ethical approval

This study does not contain any studies with human participants performed by any of the authors.

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Data will be made available on request from the corresponding author.

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